

Limits (Part III)

Finding Limits at Jump Discontinuities

Piecewise Functions

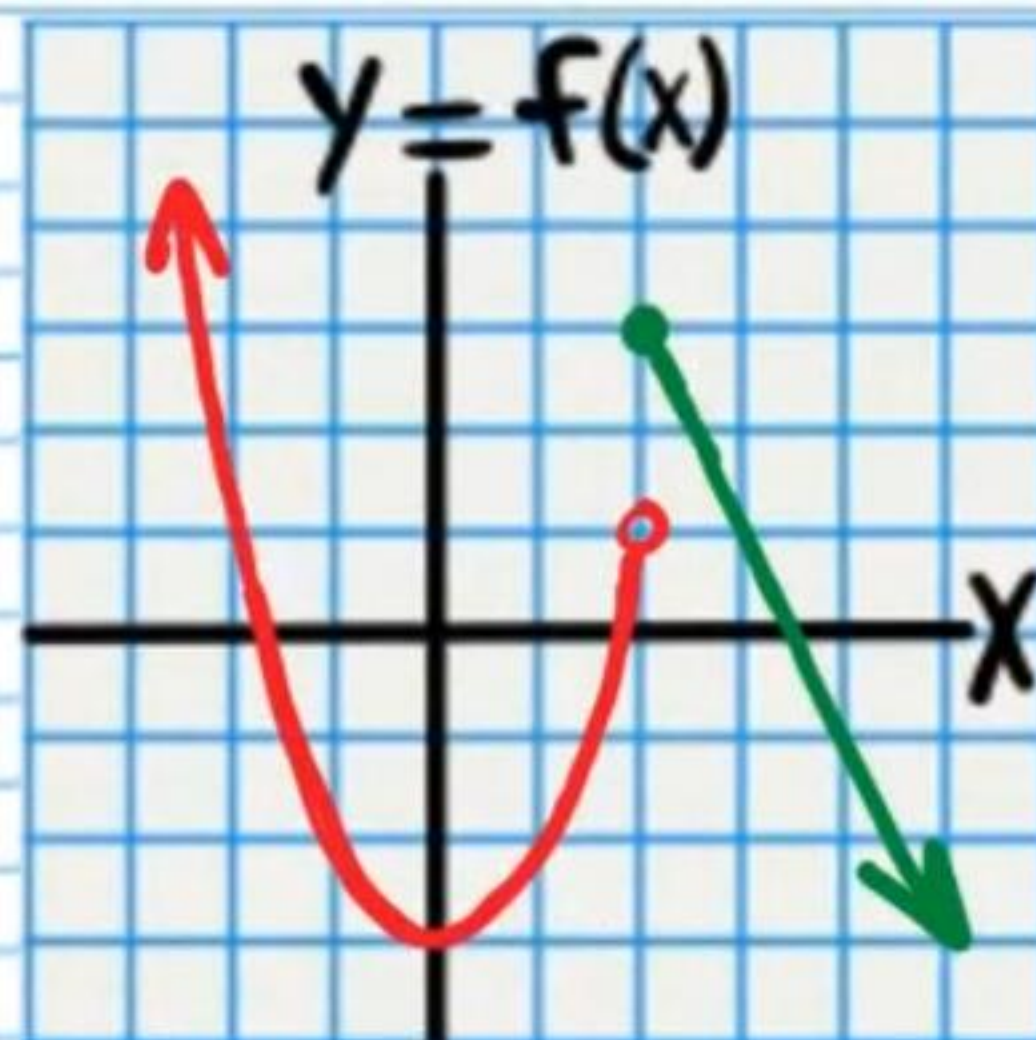
Given each function below, find the limits.

① $f(x) = \begin{cases} -2x+7, & x \geq 2 \\ x^2-3, & x < 2 \end{cases}$

i) $\lim_{x \rightarrow 2^+} f(x)$

ii) $\lim_{x \rightarrow 2^-} f(x)$

iii) $\lim_{x \rightarrow 2} f(x)$



② $f(x) = \begin{cases} \log_3(x+28), & x < -1 \\ |x+4|, & x \geq -1 \end{cases}$

i) $\lim_{x \rightarrow -1^+} f(x) = |-1+4| = |3| = \boxed{3}$

ii) $\lim_{x \rightarrow -1^-} f(x) = \log_3(-1+28) = \log_3 27 = \log_3 3^3 = \boxed{3}$

iii) $\lim_{x \rightarrow -1} f(x) = \boxed{3}$

③ $f(x) = \begin{cases} 2-x^3, & x > 3 \\ 1/x, & x \leq 3 \end{cases}$

i) $\lim_{x \rightarrow 3^+} f(x) = 2-3^3 = 2-27 = \boxed{-25}$

ii) $\lim_{x \rightarrow 3^-} f(x) = \boxed{\frac{1}{3}}$

iii) $\lim_{x \rightarrow 3} f(x) = \boxed{\text{DNE}}$

④ $f(x) = \begin{cases} 4^x+2, & x \geq -2 \\ -2(x+1)+\frac{1}{16}, & x < -2 \end{cases}$

i) $\lim_{x \rightarrow -2^+} f(x) = 4^{-2}+2 = \frac{1}{4^2}+2 = \frac{1}{16}+2 = \boxed{2\frac{1}{16}}$

ii) $\lim_{x \rightarrow -2^-} f(x) = -2(-2+1)+\frac{1}{16} = -2(-1)+\frac{1}{16} = 2+\frac{1}{16} = \boxed{2\frac{1}{16}}$

iii) $\lim_{x \rightarrow -2} f(x) = \boxed{2\frac{1}{16}}$

$$⑤ \quad f(x) = \begin{cases} x^2 + 4, & x \leq 0 \\ |2x| + 2, & x > 0 \end{cases}$$

$$i) \lim_{x \rightarrow 0^+} f(x) = |2(0)| + 2 = |0| + 2 = 0 + 2 = \boxed{2}$$

$$ii) \lim_{x \rightarrow 0^-} f(x) = 0^2 + 4 = \boxed{4}$$

$$iii) \lim_{x \rightarrow 0} f(x) = \boxed{\text{DNE}}$$

$$⑥ \quad f(x) = \begin{cases} \log_2 x, & x < 16 \\ \frac{4}{65-4x}, & x \geq 16 \end{cases}$$

$$i) \lim_{x \rightarrow 16^+} f(x) = \frac{4}{65-4(16)} = \frac{4}{65-64} = \frac{4}{1} = \boxed{4}$$

$$ii) \lim_{x \rightarrow 16^-} f(x) = \log_2 16 = \log_2 2^4 = \boxed{4}$$

$$iii) \lim_{x \rightarrow 16} f(x) = \boxed{4}$$

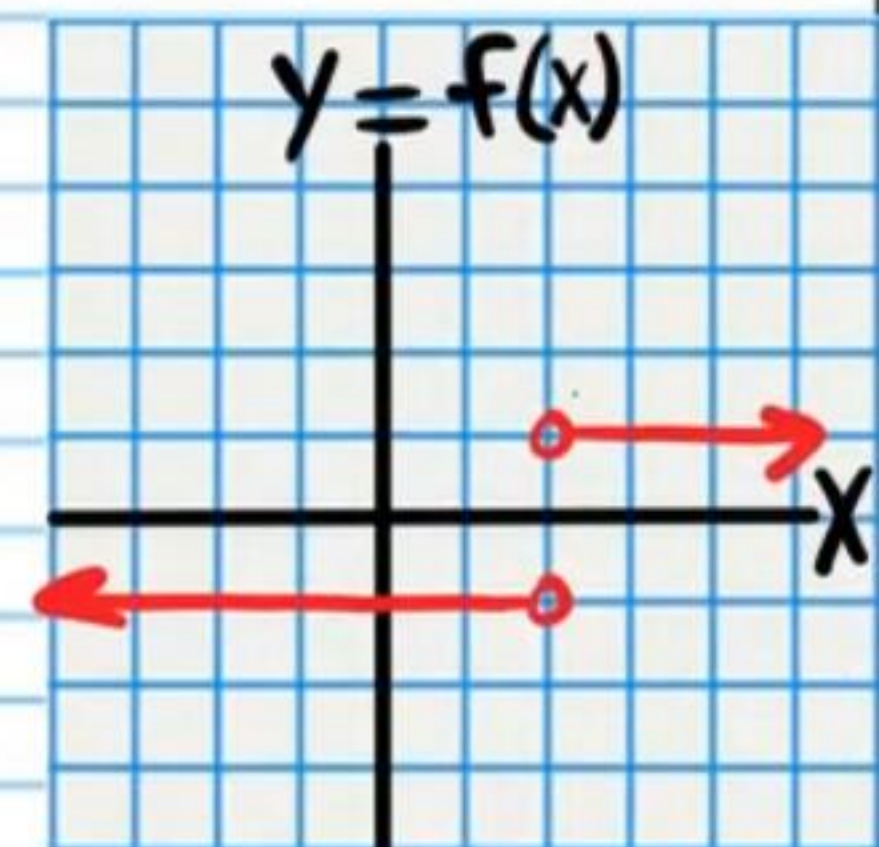
Functions of the Form $\frac{|x-a|}{x-a}$ or $\frac{x-a}{|x-a|}$

Find the limits.

$$\bullet \quad ⑦ \quad i) \lim_{x \rightarrow 2^+} \frac{|x-2|}{x-2} = \boxed{1}$$

$$ii) \lim_{x \rightarrow 2^-} \frac{|x-2|}{x-2} = \boxed{-1}$$

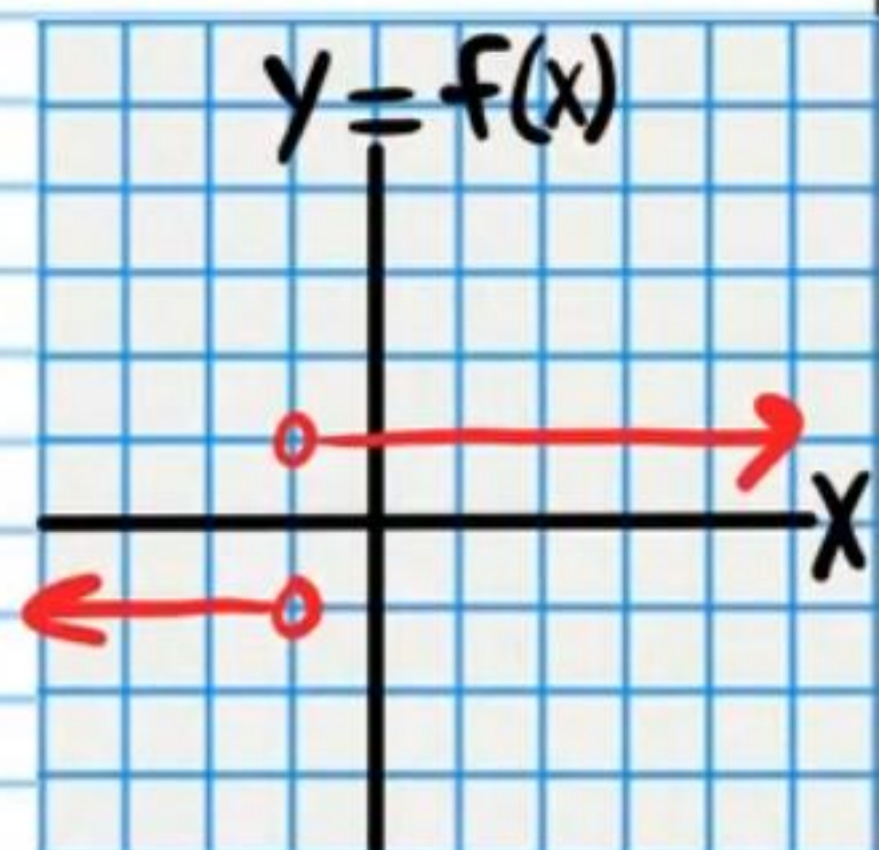
$$iii) \lim_{x \rightarrow 2} \frac{|x-2|}{x-2} = \boxed{\text{DNE}}$$



$$⑧ \quad i) \lim_{x \rightarrow -1^+} \frac{x+1}{|x+1|} = \boxed{1}$$

$$ii) \lim_{x \rightarrow -1^-} \frac{x+1}{|x+1|} = \boxed{-1}$$

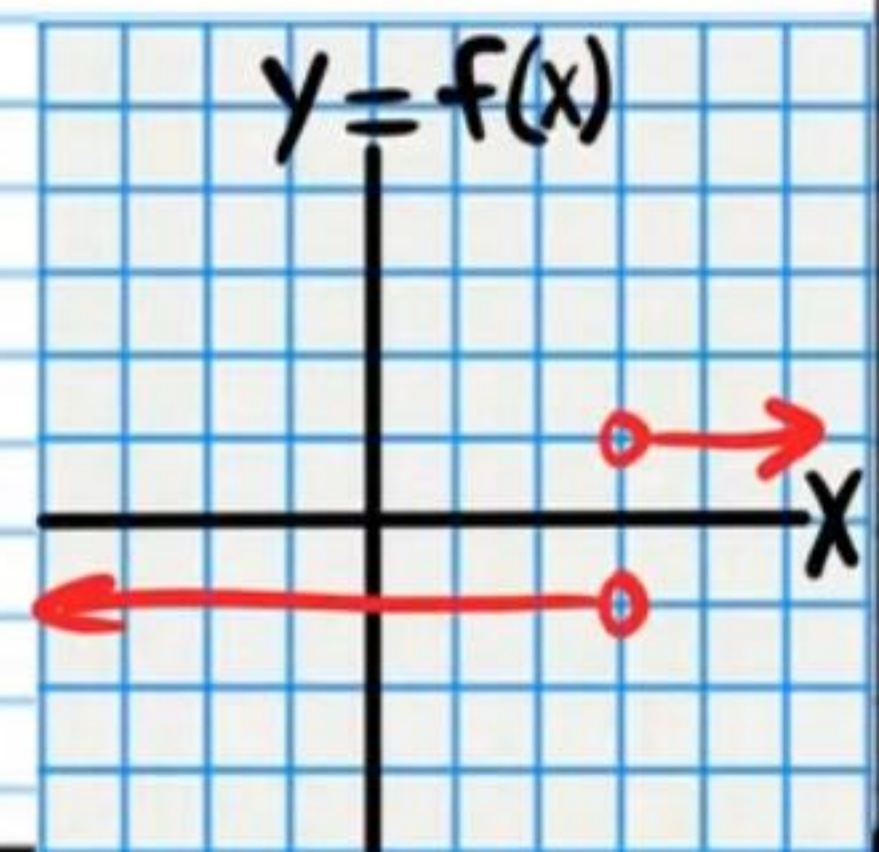
$$iii) \lim_{x \rightarrow -1} \frac{x+1}{|x+1|} = \boxed{\text{DNE}}$$



$$\bullet \quad ⑨ \quad i) \lim_{x \rightarrow 3^+} \frac{|x-3|}{x-3} = \boxed{1}$$

$$ii) \lim_{x \rightarrow 3^-} \frac{|x-3|}{x-3} = \boxed{-1}$$

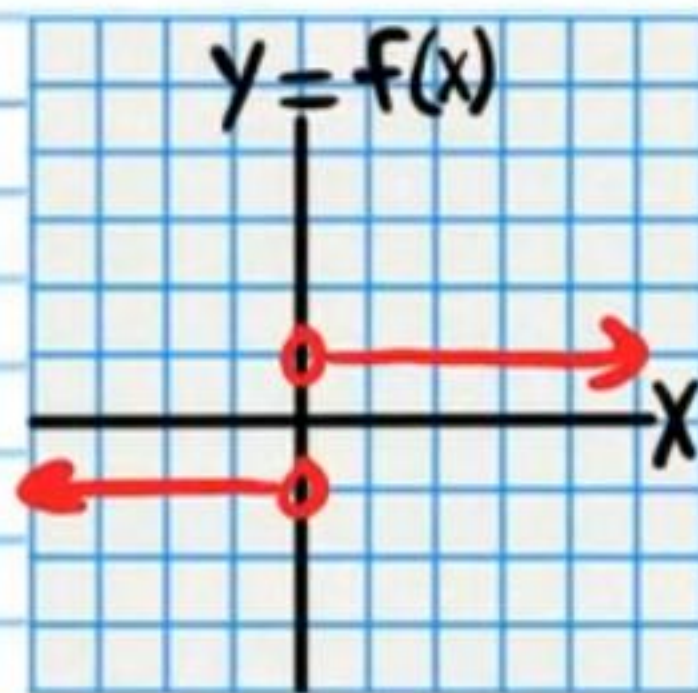
$$iii) \lim_{x \rightarrow 3} \frac{|x-3|}{x-3} = \boxed{\text{DNE}}$$



⑩ Find i) $\lim_{x \rightarrow 0^+} \frac{|x|}{x} = \boxed{1}$

ii) $\lim_{x \rightarrow 0^-} \frac{|x|}{x} = \boxed{-1}$

iii) $\lim_{x \rightarrow 0} \frac{|x|}{x} = \boxed{\text{DNE}}$



Special Limits

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

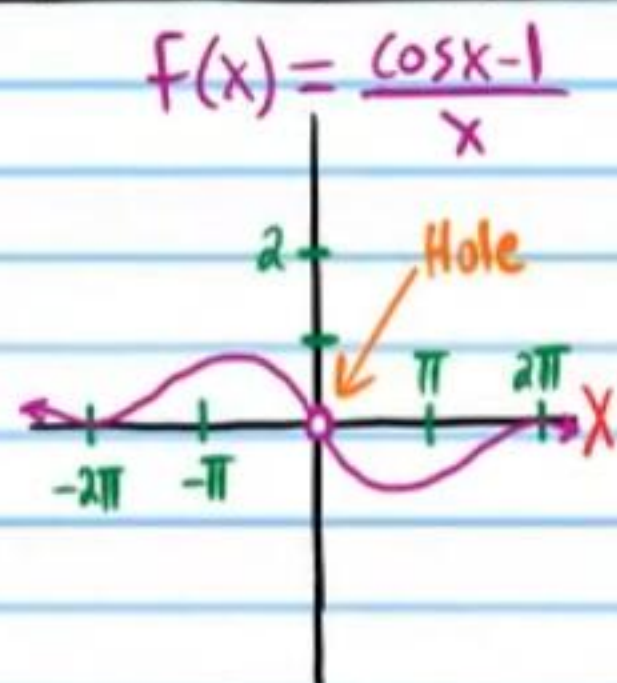
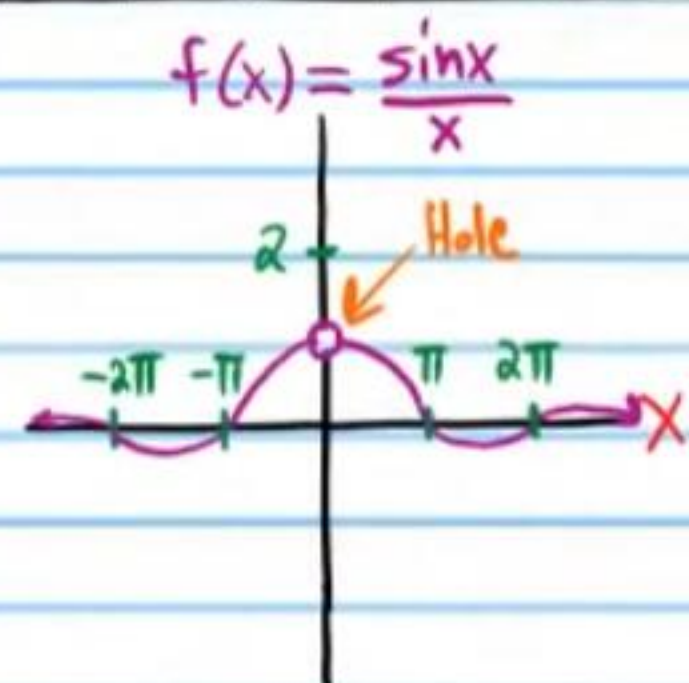
$$\lim_{x \rightarrow 0} \frac{\cos x - 1}{x} = 0$$

Demonstrating with a Table

x	0.2	0.1	0.01	0.001
f(x)	0.9933	0.998	0.9999	0.99999

x	0.2	0.1	0.01	0.001
f(x)	-0.099	-0.049	-0.0049	-0.00049

Demonstrating with a graph.



Example I) Find $\lim_{x \rightarrow 0} \frac{\sin x}{12x}$

Example II) Find $\lim_{x \rightarrow 0} \frac{x}{\cos x - 1}$

The Limit Laws

If both $\lim_{x \rightarrow a} f(x)$ and $\lim_{x \rightarrow a} g(x)$ exist, and "c" is a constant, then

1) $\lim_{x \rightarrow a} [f(x) + g(x)] = \lim_{x \rightarrow a} f(x) + \lim_{x \rightarrow a} g(x)$

2) $\lim_{x \rightarrow a} [f(x) - g(x)] = \lim_{x \rightarrow a} f(x) - \lim_{x \rightarrow a} g(x)$

3) $\lim_{x \rightarrow a} [f(x) \cdot g(x)] = \lim_{x \rightarrow a} f(x) \cdot \lim_{x \rightarrow a} g(x)$

4) $\lim_{x \rightarrow a} \left[\frac{f(x)}{g(x)} \right] = \frac{\lim_{x \rightarrow a} f(x)}{\lim_{x \rightarrow a} g(x)}, \lim_{x \rightarrow a} g(x) \neq 0$

5) $\lim_{x \rightarrow a} [c f(x)] = c \lim_{x \rightarrow a} f(x)$

6) $\lim_{x \rightarrow a} c = c$

7) $\lim_{x \rightarrow a} f(g(x)) = f\left(\lim_{x \rightarrow a} g(x)\right)$ if "f" is continuous where its input is $\lim_{x \rightarrow a} g(x)$.

8) If "f" is continuous at $x=a$, then $\lim_{x \rightarrow a} f(x) = f(a)$.
*Remember, this is the definition of continuity at $x=a$.

Find the value of each limit. Write the number of the limit law used on each step.

$$\bullet \textcircled{11} \lim_{x \rightarrow 0} \left[\frac{\sin x}{x} + \frac{\cos x - 1}{x} \right] \stackrel{1}{=} \lim_{x \rightarrow 0} \frac{\sin x}{x} + \lim_{x \rightarrow 0} \frac{\cos x - 1}{x} = 1 + 0 = \boxed{1}$$

$$\begin{aligned} \textcircled{12} \lim_{x \rightarrow 0} \left[x^2 + \frac{1 - \cos x}{x} \right] &= \lim_{x \rightarrow 0} \left[x^2 - \frac{-1 + \cos x}{x} \right] \\ &= \lim_{x \rightarrow 0} \left[x^2 - \frac{\cos x - 1}{x} \right] \\ &\stackrel{2}{=} \lim_{x \rightarrow 0} x^2 - \lim_{x \rightarrow 0} \frac{\cos x - 1}{x} \\ &\stackrel{8}{=} 0^2 - 0 \\ &= \boxed{0} \end{aligned}$$

$$\textcircled{13} \lim_{x \rightarrow 0} \frac{\sin^2 x}{x} = \lim_{x \rightarrow 0} \left[\sin x \cdot \frac{\sin x}{x} \right] \stackrel{3}{=} \lim_{x \rightarrow 0} \sin x \cdot \lim_{x \rightarrow 0} \frac{\sin x}{x}$$

$$\stackrel{8}{=} \sin(0) (1)$$

$$= (0)(1)$$

$$= \boxed{0}$$

$$\textcircled{14} \lim_{x \rightarrow 0} \frac{\cos x}{\frac{\sin x}{x}} \stackrel{4}{=} \frac{\lim_{x \rightarrow 0} \cos x}{\lim_{x \rightarrow 0} \frac{\sin x}{x}} \stackrel{8}{=} \frac{\cos(0)}{1} = \frac{1}{1} = \boxed{1}$$

$$\bullet \textcircled{15} \lim_{x \rightarrow 0} \left[\frac{\cos x - 1}{x} + \cos x \right] \stackrel{1}{=} \lim_{x \rightarrow 0} \frac{\cos x - 1}{x} + \lim_{x \rightarrow 0} \cos x \stackrel{8}{=} 0 + \cos(0) = 0 + 1 = \boxed{1}$$

$$\begin{aligned} \textcircled{16} \lim_{x \rightarrow 0} \left[\frac{\sin x}{x} - 8e^x \right] &\stackrel{2}{=} \lim_{x \rightarrow 0} \frac{\sin x}{x} - \lim_{x \rightarrow 0} 8e^x \stackrel{8}{=} 1 - 8e^0 = 1 - 8(1) \\ &= 1 - 8 \\ &= \boxed{-7} \end{aligned}$$

$$\begin{aligned} \textcircled{17} \lim_{x \rightarrow 0} \left[\frac{\sec x (1 - \cos x)}{x} - 7x \right] &\stackrel{2}{=} \lim_{x \rightarrow 0} \left[\frac{\sec x (1 - \cos x)}{x} \right] - \lim_{x \rightarrow 0} (7x) \\ &\stackrel{8}{=} \lim_{x \rightarrow 0} \left[\frac{\sec x}{1} \cdot \frac{1 - \cos x}{x} \right] - 7(0) \\ &\stackrel{3}{=} \lim_{x \rightarrow 0} \sec x \cdot \lim_{x \rightarrow 0} \frac{1 - \cos x}{x} - 0 \\ &\stackrel{8}{=} \sec(0) (0) - 0 \\ &= (1)(0) - 0 \\ &= \boxed{0} \end{aligned}$$

$$\textcircled{18} \lim_{x \rightarrow 0} \frac{1 - \cos x}{e^x} \stackrel{4}{=} \frac{\lim_{x \rightarrow 0} \frac{1 - \cos x}{x}}{\lim_{x \rightarrow 0} e^x} \stackrel{8}{=} \frac{0}{e^0} = \frac{0}{1} = \boxed{0}$$

$$\bullet (19) \lim_{x \rightarrow 0} \frac{\cos x - 1}{4x} = \lim_{x \rightarrow 0} \left[\frac{1}{4} \cdot \frac{\cos x - 1}{x} \right] \stackrel{5}{=} \frac{1}{4} \lim_{x \rightarrow 0} \frac{\cos x - 1}{x} = \frac{1}{4}(0) = \boxed{0}$$

$$(20) \lim_{x \rightarrow 0} \frac{3 \sin x}{x} = \lim_{x \rightarrow 0} \left[\frac{3}{1} \cdot \frac{\sin x}{x} \right] \stackrel{5}{=} 3 \lim_{x \rightarrow 0} \frac{\sin x}{x} = 3(1) = \boxed{3}$$

$$(21) \lim_{x \rightarrow 3} \pi \stackrel{6}{=} \boxed{\pi}$$

$$(22) \lim_{x \rightarrow 0} e^{\left(\frac{1 - \cos x}{x}\right)} \stackrel{7}{=} e^{\lim_{x \rightarrow 0} \frac{1 - \cos x}{x}} = e^0 = \boxed{1}$$

$$\bullet (23) \lim_{x \rightarrow 0} \frac{\sin x}{-2x} = \lim_{x \rightarrow 0} \left[-\frac{1}{2} \cdot \frac{\sin x}{x} \right] \stackrel{5}{=} -\frac{1}{2} \lim_{x \rightarrow 0} \frac{\sin x}{x} = -\frac{1}{2}(1) = \boxed{-\frac{1}{2}}$$

$$(24) \lim_{x \rightarrow 0} \frac{\pi \cos x - \pi}{x} = \lim_{x \rightarrow 0} \left[\frac{\pi(\cos x - 1)}{x} \right] \stackrel{5}{=} \pi \lim_{x \rightarrow 0} \frac{\cos x - 1}{x} = \pi(0) = \boxed{0}$$

$$(25) \lim_{x \rightarrow 100} 82 \stackrel{6}{=} \boxed{82}$$

* You may leave your answer in cosine form.

$$(26) \lim_{x \rightarrow 0} \cos\left(\frac{\sin x}{x}\right) \stackrel{7}{=} \cos\left(\lim_{x \rightarrow 0} \frac{\sin x}{x}\right) = \boxed{\cos(1)}$$

Challenge Problems

$$\text{Example III)} \lim_{x \rightarrow 0} \frac{x}{\sin x} = \lim_{x \rightarrow 0} \left[\frac{1}{\frac{\sin x}{x}} \right] \stackrel{4}{=} \frac{\lim_{x \rightarrow 0} 1}{\lim_{x \rightarrow 0} \frac{\sin x}{x}} \stackrel{V1}{=} \frac{1}{1} = \boxed{1}$$

$$\text{Example IV)} \lim_{x \rightarrow 0} \frac{\tan(5x)}{x} = \lim_{x \rightarrow 0} \frac{\frac{\sin(5x)}{\cos(5x)}}{x} = \lim_{x \rightarrow 0} \left[\frac{\sin(5x)}{\cos(5x)} \div x \right]$$

$$= \lim_{x \rightarrow 0} \left[\frac{\sin(5x)}{\cos(5x)} \cdot \frac{1}{x} \right]$$

$$= \lim_{x \rightarrow 0} \left[\frac{1}{\cos(5x)} \cdot \frac{\sin(5x)}{x} \right]$$



For $\lim_{x \rightarrow 0} \frac{\sin cx}{cx}$, $c = \text{constant}$, let $\theta = cx$,

$$\text{then } \lim_{x \rightarrow 0} \frac{\sin cx}{cx} = \lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1$$

$$= \lim_{x \rightarrow 0} \left[\sec(5x) \cdot \frac{\sin(5x)}{x} \right]$$

$$\stackrel{3}{=} \lim_{x \rightarrow 0} \sec(5x) \lim_{x \rightarrow 0} \left[\frac{\sin(5x)}{x} \right]$$

$$\stackrel{8}{=} \sec(5(0)) \lim_{x \rightarrow 0} \left[\frac{5 \cdot \sin(5x)}{5x} \right]$$

$$= \sec(0) \lim_{x \rightarrow 0} \left[\frac{5 \cdot \sin(5x)}{5x} \right]$$

$$\stackrel{5}{=} (1)(5) \lim_{x \rightarrow 0} \frac{\sin(5x)}{5x}$$

$$\stackrel{6}{=} (1)(5)(1)$$

$$= \boxed{5}$$

You may try these, but they are not required at this point in the course.

$$\lim_{x \rightarrow 0} \frac{1 - \cos x}{\sin(4x)}$$

Ans: $\boxed{0}$

$$\lim_{x \rightarrow 0} \frac{\sin(6x)}{\sin(5x)}$$

Ans: $\boxed{\frac{6}{5}}$

$$\lim_{x \rightarrow 0} \frac{x}{\cot x} \quad \text{Ans: } \boxed{0}$$

More Problems Requiring the Limit Laws

If $\lim_{x \rightarrow a} f(x) = -4$, $\lim_{x \rightarrow a} g(x) = 2$, and $\lim_{x \rightarrow a} h(x) = -3$, find the value of each expression. Write the number of the limit law used on each step.

• (27) $\lim_{x \rightarrow a} [f(x)g(x)] \stackrel{3}{=} \lim_{x \rightarrow a} f(x) \lim_{x \rightarrow a} g(x) = -4(2) = \boxed{-8}$

(28) $\lim_{x \rightarrow a} \left[\frac{g(x)}{h(x)} \right] \stackrel{4}{=} \frac{\lim_{x \rightarrow a} g(x)}{\lim_{x \rightarrow a} h(x)} = \frac{2}{-3} = \boxed{-\frac{2}{3}}$

(29) $\lim_{x \rightarrow a} [h(x) + 6g(x)] \stackrel{1}{=} \lim_{x \rightarrow a} h(x) + \lim_{x \rightarrow a} [6g(x)] \stackrel{5}{=} -3 + 6 \lim_{x \rightarrow a} g(x)$
 $= -3 + 6(2)$
 $= \boxed{9}$

$\lim_{x \rightarrow a} f(x) = -4$, $\lim_{x \rightarrow a} g(x) = 2$, and $\lim_{x \rightarrow a} h(x) = -3$.

(30) $\lim_{x \rightarrow a} [4 - h(x)] \stackrel{2}{=} \lim_{x \rightarrow a} 4 - \lim_{x \rightarrow a} h(x) \stackrel{6}{=} 4 - (-3) = \boxed{7}$

• (31) $\lim_{x \rightarrow a} [g(x) - f(x)] \stackrel{2}{=} \lim_{x \rightarrow a} g(x) - \lim_{x \rightarrow a} f(x) = 2 - (-4) = \boxed{6}$

(32) $\lim_{x \rightarrow a} \left[\frac{h(x)}{4f(x)} \right] \stackrel{4}{=} \frac{\lim_{x \rightarrow a} h(x)}{\lim_{x \rightarrow a} [4f(x)]} \stackrel{5}{=} \frac{-3}{4 \lim_{x \rightarrow a} f(x)} = \frac{-3}{4(-4)} = \boxed{\frac{3}{16}}$

$\lim_{x \rightarrow a} f(x) = -4$, $\lim_{x \rightarrow a} g(x) = 2$, and $\lim_{x \rightarrow a} h(x) = -3$.

(33) $\lim_{x \rightarrow a} [g(x)h(x)] \stackrel{3}{=} \lim_{x \rightarrow a} g(x) \lim_{x \rightarrow a} h(x) = 2(-3) = \boxed{-6}$

(34) $\lim_{x \rightarrow a} [-5 + f(x)] \stackrel{1}{=} \lim_{x \rightarrow a} (-5) + \lim_{x \rightarrow a} f(x) \stackrel{6}{=} -5 - 4 = \boxed{-9}$

• (35) $\lim_{x \rightarrow a} [g(x)]^2 \stackrel{7}{=} (\lim_{x \rightarrow a} g(x))^2 = (2)^2 = \boxed{4}$

(36) $\lim_{x \rightarrow a} \cos(h(x)) \stackrel{7}{=} \cos(\lim_{x \rightarrow a} h(x)) = \cos(-3) = \boxed{\cos 3}$

$\lim_{x \rightarrow a} f(x) = -4$, $\lim_{x \rightarrow a} g(x) = 2$, and $\lim_{x \rightarrow a} h(x) = -3$.

(37) $\lim_{x \rightarrow a} [x^2 - 3x + 1] \stackrel{8}{=} a^2 - 3a + 1$

• (38) $\lim_{x \rightarrow a} \sqrt[3]{f(x)} \stackrel{7}{=} \sqrt[3]{\lim_{x \rightarrow a} f(x)} = \sqrt[3]{-4} = \boxed{-\sqrt[3]{4}}$

(39) $\lim_{x \rightarrow a} e^{f(x)} \stackrel{7}{=} e^{\lim_{x \rightarrow a} f(x)} = e^{-4} = \boxed{\frac{1}{e^4}}$

(40) $\lim_{x \rightarrow a} (2x - 5) \stackrel{8}{=} 2a - 5$

$$\lim_{x \rightarrow a} f(x) = -4, \lim_{x \rightarrow a} g(x) = 2, \text{ and } \lim_{x \rightarrow a} h(x) = -3.$$

$$\bullet \textcircled{41} \lim_{x \rightarrow a} \frac{1}{g(x)-4} \stackrel{4}{=} \frac{\lim_{x \rightarrow a} 1}{\lim_{x \rightarrow a} [g(x)-4]} \stackrel{2 \& 6}{=} \frac{1}{\lim_{x \rightarrow a} g(x) - \lim_{x \rightarrow a} 4} \stackrel{6}{=} \frac{1}{2-4} = \boxed{-\frac{1}{2}}$$

$$\textcircled{42} \lim_{x \rightarrow a} [x h(x)] \stackrel{3}{=} \lim_{x \rightarrow a} x \lim_{x \rightarrow a} h(x) \stackrel{8}{=} a(-3) = \boxed{-3a}$$

$$\bullet \textcircled{43} \lim_{x \rightarrow a} (2+h(x))^2 \stackrel{7}{=} \left(\lim_{x \rightarrow a} [2+h(x)] \right)^2 \stackrel{1}{=} \left(\lim_{x \rightarrow a} 2 + \lim_{x \rightarrow a} h(x) \right)^2 \\ \stackrel{6}{=} (2-3)^2 \\ = (-1)^2 \\ = \boxed{1}$$

$$\textcircled{44} \lim_{x \rightarrow a} \frac{x}{2f(x)} \stackrel{4}{=} \frac{\lim_{x \rightarrow a} x}{\lim_{x \rightarrow a} [2f(x)]} \stackrel{5 \& 8}{=} \frac{a}{2 \lim_{x \rightarrow a} f(x)} = \frac{a}{2(-4)} = \boxed{-\frac{a}{8}}$$

Other Types of Limits at Removable Discontinuities

Adding/Subtracting Fractions

Find the values of the limits algebraically.

$$\bullet \textcircled{45} \lim_{x \rightarrow -\frac{5}{2}} \left[\frac{1}{2x+5} + \frac{x+3}{2x^2+9x+10} \right] = \lim_{x \rightarrow -\frac{5}{2}} \left[\frac{1}{2x+5} + \frac{x+3}{(2x+5)(x+2)} \right] \\ = \lim_{x \rightarrow -\frac{5}{2}} \left[\frac{1}{(2x+5)\cancel{(x+2)}} + \frac{x+3}{(2x+5)(x+2)} \right] \\ = \lim_{x \rightarrow -\frac{5}{2}} \left[\frac{x+2}{(2x+5)(x+2)} + \frac{x+3}{(2x+5)(x+2)} \right] \\ = \lim_{x \rightarrow -\frac{5}{2}} \left[\frac{2x+5}{(2x+5)(x+2)} \right] \\ = \lim_{x \rightarrow -\frac{5}{2}} \frac{1}{x+2} \\ = \frac{1}{-\frac{5}{2}+2} \\ = \frac{1}{-\frac{1}{2}} \\ = 1 \div \left(-\frac{1}{2}\right) = \frac{1}{1} \cdot \left(-\frac{2}{1}\right) = \boxed{-2}$$

$$\textcircled{46} \lim_{b \rightarrow -2} \left[\frac{2b}{b^2-4} - \frac{1}{b+2} \right] = \lim_{b \rightarrow -2} \left[\frac{2b}{(b-2)(b+2)} - \frac{1}{b+2} \right]$$

$$= \lim_{b \rightarrow -2} \left[\frac{2b}{(b-2)(b+2)} - \frac{1 \cdot (b-2)}{(b+2)(b-2)} \right]$$

$$= \lim_{b \rightarrow -2} \left[\frac{2b}{(b-2)(b+2)} - \frac{b-2}{(b+2)(b-2)} \right]$$

$$= \lim_{b \rightarrow -2} \left[\frac{b+2}{(b-2)(b+2)} \right]$$

$$= \lim_{b \rightarrow -2} \frac{1}{b-2}$$

$$= \frac{1}{-2-2}$$

$$= \boxed{-\frac{1}{4}}$$

$$\textcircled{47} \lim_{t \rightarrow 0} \left[\frac{1}{t-3} + \frac{1}{t} + \frac{3}{t^2-3t} \right] = \lim_{t \rightarrow 0} \left[\frac{1}{t-3} + \frac{1}{t} + \frac{3}{t(t-3)} \right]$$

$$= \lim_{t \rightarrow 0} \left[\frac{1 \cdot (t)}{(t-3)(t)} + \frac{1 \cdot (t-3)}{t(t-3)} + \frac{3}{t(t-3)} \right]$$

$$= \lim_{t \rightarrow 0} \left[\frac{t}{t(t-3)} + \frac{t-3}{t(t-3)} + \frac{3}{t(t-3)} \right]$$

$$= \lim_{t \rightarrow 0} \left[\frac{2t}{t(t-3)} \right]$$

$$= \lim_{t \rightarrow 0} \frac{2}{t-3}$$

$$= \frac{2}{0-3}$$

$$= \boxed{-\frac{2}{3}}$$

$$\begin{aligned}
 \textcircled{48} \lim_{x \rightarrow 1} \left[\frac{4x+1}{3x^2-x-2} - \frac{1}{x-1} \right] &= \lim_{x \rightarrow 1} \left[\frac{4x+1}{(3x+2)(x-1)} - \frac{1}{x-1} \right] \\
 &= \lim_{x \rightarrow 1} \left[\frac{4x+1}{(3x+2)(x-1)} - \frac{1 \cdot (3x+2)}{(x-1)(3x+2)} \right] \\
 &= \lim_{x \rightarrow 1} \left[\frac{4x+1}{(3x+2)(x-1)} - \frac{3x+2}{(x-1)(3x+2)} \right] \\
 &= \lim_{x \rightarrow 1} \left[\frac{x-1}{(3x+2)(x-1)} \right] \\
 &= \lim_{x \rightarrow 1} \frac{1}{3x+2} \\
 &= \frac{1}{3(1)+2} \\
 &= \boxed{\frac{1}{5}}
 \end{aligned}$$

$$\textcircled{49} \lim_{r \rightarrow \frac{7}{2}} \left[\frac{1}{2r-7} + \frac{2r-21}{4r^2-49} - \frac{1}{2r+7} \right]$$

$$\lim_{r \rightarrow \frac{7}{2}} \left[\frac{1}{2r-7} + \frac{2r-21}{(2r+7)(2r-7)} - \frac{1}{2r+7} \right]$$

$$\lim_{r \rightarrow \frac{7}{2}} \left[\frac{1 \cdot (2r+7)}{(2r-7)(2r+7)} + \frac{2r-21}{(2r+7)(2r-7)} - \frac{1 \cdot (2r-7)}{(2r+7)(2r-7)} \right]$$

$$\lim_{r \rightarrow \frac{7}{2}} \left[\frac{2r+7}{(2r-7)(2r+7)} + \frac{2r-21}{(2r+7)(2r-7)} - \frac{2r-7}{(2r+7)(2r-7)} \right]$$

$$\lim_{r \rightarrow \frac{7}{2}} \left[\frac{2r-7}{(2r-7)(2r+7)} \right]$$

$$\lim_{r \rightarrow \frac{7}{2}} \frac{1}{2r+7}$$

$$\frac{1}{2(\frac{7}{2})+7}$$

$$\frac{1}{7+7}$$

$$\boxed{\frac{1}{14}}$$

$$\textcircled{50} \lim_{m \rightarrow 0} \left[\frac{3}{m} - \frac{2m+18}{m^2+6m} \right] = \lim_{m \rightarrow 0} \left[\frac{3}{m} - \frac{2m+18}{m(m+6)} \right]$$

$$= \lim_{m \rightarrow 0} \left[\frac{3(m+6)}{m(m+6)} - \frac{2m+18}{m(m+6)} \right]$$

$$= \lim_{m \rightarrow 0} \left[\frac{3m+18}{m(m+6)} - \frac{2m+18}{m(m+6)} \right]$$

$$= \lim_{m \rightarrow 0} \left[\frac{m}{m(m+6)} \right]$$

$$= \lim_{m \rightarrow 0} \frac{1}{m+6}$$

$$= \frac{1}{0+6}$$

$$= \boxed{\frac{1}{6}}$$